A Low-Cost and Low-Complexity Motion Sensor System Using IMU and Video-Based Tracking

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Abstract

This report presents MoveSync, a low-cost, browser-based motion analysis prototype developed for sports performance analysis. The system integrates two complementary sensing modalities, video-based pose estimation and inertial measurement unit (IMU) sensors, within an open-source, client-side web application requiring no downloads or server infrastructure. Video pose estimation is performed using MoveNet via TensorFlow.js, extracting two-dimensional skeletal keypoints from uploaded recordings and displaying them in real time. IMU sensor data is processed through a pipeline comprising Madgwick sensor fusion, gravity removal, and zero-velocity update (ZUPT)-based integration to derive motion metrics including peak acceleration, peak velocity, and total distance. A manual event-based synchronisation method aligns the two data streams, enabling coherent visualisation and inspection of multimodal motion data. The application is structured around a project–session data model and has six main functional modules: Dashboard, Sport Presets, Upload, Library, Session Viewer, and Compare Sessions. A controlled validation experiment evaluated the accuracy of IMU-derived velocity and distance estimates across ten 30 cm horizontal movements. The results revealed substantial overestimation attributable to cumulative integration drift, with quick movements overestimating distance by a mean factor of 5.3 and slower movements by 67.9. These findings highlight the fundamental limitations of double integration from MEMS IMUs and identify signal processing improvement as the primary direction for future work. Despite these limitations, MoveSync demonstrates the feasibility of accessible, browser-based multimodal motion capture and establishes a functional foundation for further development.

Contributions

The development of the MoveSync system has been a collaborative effort, with overall contributions shared equally among group members. This includes the design and implementation of the web application codebase, the preparation and delivery of the group presentations, and the writing of all reports. All members attended weekly meetings with the project supervisor as well as additional group meetings when necessary, contributing equally during these.

Jack Hutton

Jack Hutton primarily worked on the IMU pipeline for this project. This involved modifying the Session Viewer page, implementing sensor fusion and 3D orientation visualisations, creating the ZUPT detection, and improving the filtering and gravity removal. He also implemented the sports metrics into the web application, modifying the Sports Preset page and Session Viewer significantly to calculate and display them. Throughout the semester, he contributed numerous fixes across different features and sections as problems arose and bugs were found. For this report, he was responsible for Sections 2.3, 2.5, 3.4, 3.6, 4.2, 5.2, and the Abstract, and contributed to the overall report through planning, editing, and proofreading.

Alexander Jonsson

Alexander Jonsson primarily worked on developing the filtering pipeline for the IMU data in this project. This involved analyzing the trade-offs between different filtering approaches and determining where filters should be applied and where they should be avoided in order to preserve signal quality. He also assisted Jack Hutton in the development of the sensor fusion component and the implementation of the 3D visualisation of the sensor orientation. Throughout the project, he contributed extensively to debugging the system by identifying unexpected errors in the codebase, tracing their underlying causes, and proposing or implementing potential fixes when issues arose. For this report, he was responsible for Sections 2.5.1, 2.6, 3.5, and 5.3, and also contributed to the overall report through minor revisions and adjustments across several other sections.

Sahand Khodabandehloo

Sahand Khodabandehloo was primarily responsible for the design and documentation of the web application as a complete system, including its purpose, workflow, and methodological foundation. In the report, he was responsible for sections 2.1, 2.2, 2.4, 3.1, 3.2, 3.3, 4.1, 5.1, 5.3, and the Conclusion. His contributions were particularly centered on presenting the system architecture, motivating the project in relation to its application domain, and ensuring a clear and coherent description of the overall development process and resulting web application.

Contents

Abstract i

Contributions ii

1 Introduction 1
- 1.1 Motivation . . . 1
- 1.2 Aim and objectives . . . 1
- 1.3 Requirements and scope . . . 1

2 Background 3
- 2.1 Motion capture technologies . . . 3
  - 2.1.1 Current requirement constraints and trade-offs . . . 3
  - 2.1.2 Application potential of motion capture in sports fields . . . 3
- 2.2 Video pose estimation . . . 3
- 2.3 Inertial Measurement Units . . . 4
- 2.4 Synchronization . . . 4
- 2.5 Signal Processing . . . 4
  - 2.5.1 Filtering . . . 5
  - 2.5.2 Sensor Fusion . . . 5
  - 2.5.3 Gravity Removal . . . 5
  - 2.5.4 Zero-Velocity Updates . . . 6
- 2.6 Sports Analysis Metrics . . . 6
  - 2.6.1 Acceleration-based metrics . . . 6
  - 2.6.2 Velocity-distance metrics . . . 6
  - 2.6.3 Angular velocity metrics . . . 7

3 Methods 8
- 3.1 System Architecture Design Principles . . . 8
  - 3.1.1 Project–Session–Based Data Model . . . 8
  - 3.1.2 Data Stream Processing and Synchronization . . . 8
  - 3.1.3 Client-Side Execution Environment . . . 9
- 3.2 Web Application Design and Implementation . . . 9
  - 3.2.1 Application Module: Dashboard . . . 9
  - 3.2.2 Application Module: Sport Presets . . . 9
  - 3.2.3 Application Module: Upload . . . 9
  - 3.2.4 Application Module: Library . . . 9
  - 3.2.5 Application Module: Session Viewer . . . 10
  - 3.2.6 Application Module: Compare Sessions . . . 10
- 3.3 Synchronization . . . 10
- 3.4 IMU Data Processing Pipeline . . . 10
  - 3.4.1 Pre-processing and Sensor Calibration . . . 10
  - 3.4.2 Sensor Fusion . . . 10
  - 3.4.3 Velocity and Displacement from ZUPT Integration . . . 11

3.5 Sports Metrics Calculation 11
3.6 Speed and Distance Experiment 12

4 Results **13**
4.1 Web Application Progress 13
4.1.1 Modules 13
4.1.2 Data Management and Browser-Based Performance 14
4.2 Speed and Distance Validation 14

5 Discussion **16**
5.1 System Workflow and Usability 16
5.2 Validation Experiment Results 16
5.3 Ethics 17
5.3.1 Privacy, data protection, and informed consent 17
5.3.2 Misinterpretation and over-reliance 17
5.3.3 Third-party and open-source code 18
5.4 Future Work 18

6 Conclusion **20**

1 Introduction

1.1 Motivation

In competitive sports, small margins often determine the difference between success and failure. Improvements in time, joint angles, force application, and movement coordination can have a substantial impact on athletic activities (e.g., sprinting, jumping, throwing, and strength training). For this reason, biomechanical analysis has become an increasingly important component of modern sports science, supporting evidence-based coaching, performance optimisation, and injury prevention (1).

Despite extensive scientific knowledge of movement quality and its influence on performance and injury risk, many training decisions still rely heavily on visual assessment and subjective coaching experience rather than objective biomechanical data. Although expert observation remains valuable, it is inherently limited in precision, consistency, and reproducibility. This creates a gap between scientific methods for motion analysis and their practical use in everyday training environments (1). Therefore, there is a clear need for accessible tools that can provide quantitative movement analysis without requiring expensive laboratory equipment or highly specialised technical expertise.

Recent advances in wearable sensors, computer vision, and browser-based technologies have created new possibilities for low-cost motion capture systems. In particular, video-based pose estimation and wearable sensors offer complementary sources of biomechanical information: on the one hand, video-based pose estimation provides interpretable visual representations of body movement; on the other hand, wearable sensors capture acceleration and rotational motion (1). Integrating these technologies into a single browser-based application could enable practical and reproducible sports performance analysis in real-world settings.

1.2 Aim and objectives

The aim of this project is to design and develop a low-cost, low-complexity, browser-based motion tracking prototype for sports performance analysis. The proposed system, *MoveSync*, is intended to enable athletes and coaches to analyse recorded movements using only a consumer-grade camera and wearable IMU sensors, without requiring specialised software. The system is designed to support accessible, practical, and reproducible motion analysis in real-world training contexts.

To achieve this aim, the project was guided by the following objectives:

1. To perform browser-based video pose estimation to extract two-dimensional skeletal keypoints from recorded sports movements.
2. To develop a synchronisation method capable of aligning video frames with IMU data despite their independent time bases.
3. To provide an intuitive interface for the visualisation of synchronised video and sensor data to support performance analysis.

1.3 Requirements and scope

To fulfil its intended role in sports performance analysis, the system should satisfy several practical and technical requirements:

- It should be affordable and rely on commercially available cameras and wearable IMUs.

- It should operate entirely within a standard web browser without requiring complex installation procedures or specialised software or hardware.
- It should provide synchronised visualisation of skeletal motion and sensor signals to facilitate interpretation of movement patterns.
- It should allow event marking and annotation (e.g., jump take-off, landing, sprint start, and lift initiation) to support structured performance review.
- It should enable import of recorded video and IMU data.
- It should enable export of project data and reports for documentation and coach-athlete feedback.

These design priorities are consistent with literature emphasising in-field usability, affordable sensing technologies, and interpretable movement analysis for applied sports contexts (2, 3). Nevertheless, the scope of the project is limited: *MoveSync* is intended for exploratory performance analysis and training support rather than laboratory-grade biomechanical validation. It is designed as a prototype that prioritises accessibility, interpretability, and usability over maximal measurement precision. Accordingly, the system should NOT be used for medical diagnosis or clinical evaluation.

2 Background

2.1 Motion capture technologies

Motion capture is the process of translating physical movement in three-dimensional (3D) space into digitized, measurable information. By quantifying movement in terms of kinematic and dynamic variables (e.g., joint trajectories, segment velocities, accelerations, and event timings), the data enables high-accuracy reconstruction of a 2D or 3D model, allowing for systematic and reproducible evaluation of the performed motion (4). Also, given a range of existing systems (e.g., cinematography, electromagnetic, and computer vision capture systems), researchers and practitioners must select appropriate tools for their specific needs (5).

2.1.1 Current requirement constraints and trade-offs

The use of laboratory-based optical systems is typically restricted to controlled environments due to technical complexity and resource limitations (6). Furthermore, different systems involve trade-offs between measurement accuracy, precision, portability, robustness, and practical deployment. For instance, in comparison with wearable sensor-based systems, even though computer-vision-based pose estimation enables markerless motion capture from video inputs, its performance can be limited by camera perspective constraints, lighting conditions, and the lack of sport-specific datasets in complex scenarios (5).

2.1.2 Application potential of motion capture in sports fields

In sport performance analysis, motion capture serves a different primary objective than laboratory biomechanics research: it must support practical training decisions in dynamic, time-constrained environments based on the performed motion (i.e., a sole, captured model does not suffice) (1, 5). Applications in sports include objective assessment of technique, monitoring of performance progression and training adaptations over time, detecting asymmetries, and evaluating efficiency. Feedback from such informs technical performance-enhancing adjustments and injury-prevention strategies (1, 5). Consequently, motion capture is not only an advantageous tool for coaches and athletes but also has the potential to revolutionize the sports world. Yet, not only is access to laboratory-grade systems in everyday practice limited, but the transfer of technology-assisted movement assessment from the lab to the field is also still emerging (5, 1). As a result, there is a growing demand for motion analysis tools that balance measurement capability with usability, accessibility, and real-world deployment (5). Additionally, systems that integrate portable sensing technologies with intuitive analysis interfaces may help bridge the gap between biomechanical research and applied sports performance practice (5).

2.2 Video pose estimation

Video pose estimation is a computer-vision approach that infers a human pose from standard color (RGB) video frames by predicting a set of body keypoints for each frame, which can be visualized as a skeletal overlay (7). In this project, pose estimation was performed in the browser using TensorFlow.js models. The implementation initially used PoseNet as an early baseline, but the final system transitioned to MoveNet due to its better suitability for robust and accurate video-based pose tracking(7, 8).

MoveNet is a pose detector provided through the TensorFlow.js pose-detection API (8). It predicts 2D body keypoints and associated confidence scores from individual video frames, and is designed to provide better speed, robustness, and temporal stability across frames, reducing jitter and making predicted motion trajectories more suitable for video analysis (8). These characteristics made it well suited for the implementation in this project.

MoveNet provides two single-person variants, Lightning and Thunder, where Lightning is optimised more for speed and Thunder for higher accuracy and stability (8). Since the final system focuses on offline processing of uploaded videos rather than real-time performance, the implementation prioritises the more stable and accurate variant.

2.3 Inertial Measurement Units

Inertial Measurement Units (IMUs) are electronic devices that contain an array of sensors designed to measure motion. They typically comprise a combination of gyroscopes, accelerometers, and sometimes magnetometers to capture the rotational and translational motion of a body. The gyroscope measures angular velocity about each axis, the accelerometer measures linear acceleration along each axis, and the magnetometer provides a reference to the Earth's magnetic field for estimating absolute heading (9). Together, these sensors provide up to nine degrees of freedom (DOF) of raw motion data, which can be used to estimate the full three-dimensional orientation of a body segment (10).

Most IMU devices are micro-electro-mechanical systems (MEMS), which range in size and complexity. These devices typically operate on frequencies ranging from 100–2000 Hz depending on the sensor. MEMS IMUs are commonly used in wearable devices and consumer electronics due to their small form factors, minimal power consumption, and low production cost (10). Higher performance IMUs do exist that use fibre optic gyroscopes (FOGs) or ring laser gyroscopes (RLGs), particularly in the maritime and aerospace fields, but these are significantly more expensive and complex (10).

IMU devices are becoming increasingly common in human motion analysis for biomechanical and clinical research. This is mainly because of their ability to measure movement outside of a laboratory setting, whilst being low-cost and small. However, there are several limitations to IMU technology, particularly MEMS IMUs, that constrain their use without additional correction or processing. MEMS IMUs are susceptible to a number of error sources, including temperature-dependent drift, scale factor errors, cross-axis sensitivity, random measurement noise, and bias (11). Particularly significant issues arise as a result of noisy gyroscope and acceleration data which, when integrated over time, can cause cumulative orientation and position error. This necessitates signal processing to obtain more stable and accurate results for analysis.

2.4 Synchronization

A crucial step before qualitatively assessing the motion is to confirm that the video and IMU recordings, when combined, have the same time base. Synchronization refers to establishing a consistent mapping between the video and IMU recording timelines, so corresponding samples represent the same physical instant. A common and practical approach is event-based synchronization, which uses a distinct, clearly identifiable motion (e.g., a hand clap, a jump landing, or a sudden knee flexion) as a reference point for both signals.

Once synchronized, the IMU signals can be interpreted directly in the context of the observed motion in the video. This improves interpretability, supports clearer analysis, and reduces ambiguity during inspection and annotation.

2.5 Signal Processing

The raw data recorded by an IMU device can be used to derive several important motion metrics, but the errors inherent to MEMS sensors, such as noise, bias, and drift, must be addressed before reliable quantities can be extracted. A range of signal processing techniques can be applied to mitigate these errors, including digital filtering, sensor fusion, gravity removal, and zero-velocity updates.

2.5.1 Filtering

Digital filters are commonly applied to raw IMU signals to reduce measurement noise prior to analysis. A low-pass filter can be used to attenuate high-frequency noise while preserving the lower-frequency motion content of interest (human movements) (12). This is particularly important for metrics that are calculated by integrating acceleration. Since integration accumulates errors over time, even small residual noise can result in significant drift in estimated velocities and positions (13). While standard human motions such as walking and hand gestures generally fall below 10 Hz (14), it is reasonable to expect that more dynamic sports movements involving rapid impacts or high-speed limb motion may contain frequency components exceeding this range.

2.5.2 Sensor Fusion

When working with IMU data, sensor fusion can help reconcile the complementary error characteristics of the gyroscope, accelerometer, and magnetometer. The gyroscope provides precise short-term angular velocity measurements but accumulates drift over time, whereas the accelerometer and magnetometer provide stable long-term reference vectors that can be used to correct this drift (15). By combining these measurements, sensor fusion algorithms exploit the strengths of each sensor while compensating for their individual weaknesses.

The Madgwick algorithm, developed by Sebastian Madgwick in 2010, is a widely adopted approach for IMU sensor fusion due to its computational efficiency and suitability for real-time applications (16). It represents orientation as a quaternion and uses an analytically derived gradient-descent step to continuously correct gyroscope integration error using the accelerometer and magnetometer as stable reference vectors (16).

Quaternions are a four-component representation of orientation in three-dimensional space, expressing a rotation as a scalar part w and a vector part (x, y, z).

$$q = w + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}, \quad \text{where } w^2 + x^2 + y^2 + z^2 = 1$$

Unlike traditional Euler angles, they do not suffer from gimbal lock, a condition in which one or more rotational degrees of freedom is lost when rotation axes align. In addition, quaternion operations are computationally efficient for composing and interpolating rotations (17). In quaternion-based orientation filters, such as the Madgwick filter, the orientation estimate is updated by combining gyroscope integration with corrective feedback derived from accelerometer and magnetometer measurements. A single adjustable gain parameter, β, governs the balance between the gyroscope-based prediction and this correction term (16).

After stable orientation estimates are obtained via sensor fusion, a wide range of further analyses become possible. Orientation can be expressed as Euler angles (roll, pitch, yaw) or a rotation matrix, which can be used to create a real-time 3D visualisation of the sensor's position and orientation. Furthermore, when multiple IMUs are worn on adjacent body segments simultaneously, the relative orientation between segments can be computed from their respective quaternions or rotation matrices, enabling continuous joint angle estimation (18).

2.5.3 Gravity Removal

Raw accelerometer data contains contributions from both the inertial acceleration of the sensor and gravitational acceleration. Since gravity is a constant bias acting on all axes of the accelerometer, failing to remove it would introduce a significant and persistent error into any subsequent integration, making accurate velocity and displacement estimates impossible. To isolate the true linear acceleration of a body segment, the gravitational component must therefore be removed. Using the orientation estimate obtained from sensor fusion, the gravity vector can be rotated from the world frame into the sensor body frame using the conjugate of the

orientation quaternion and subtracted from the raw accelerometer reading. The resulting linear acceleration may then be rotated back into the world frame to maintain consistent axis orientation throughout the recording (19).

2.5.4 Zero-Velocity Updates

When integrating acceleration to derive kinematic quantities such as velocity and displacement, residual errors in the acceleration signal cause the integrated velocity to drift over time, a problem that compounds with recording duration (13). Zero-velocity updates (ZUPT) are a widely used technique to bound this drift by detecting periods during which the sensor is stationary and resetting the integrated velocity to zero at those points (20). A stationary sensor can be identified by the absence of meaningful angular velocity and a total accelerometer magnitude close to 1 g, reflecting gravity as the only acting force.

2.6 Sports Analysis Metrics

Quantitative metrics are fundamental in sports analysis because they provide an objective way to evaluate athlete performance, reducing the reliance on subjective judgment. In this section, several important metrics will be discussed, what they are, and their relevance for sport science. The metrics can be divided into three groups: acceleration-based metrics, velocity-distance metrics, and angular velocity metrics.

2.6.1 Acceleration-based metrics

Peak acceleration is a metric defined as the highest acceleration measured during a session. It is commonly measured with the standard gravity unit g, where 1 g equals approximately 9.8 m/s^2. In sports science, peak acceleration can be used to assess injury risk during high-impact movements, such as landings or sprint foot strikes, because it is related to mechanical loading rates. For example, runners exhibiting axial peak tibial acceleration above 9 g during overground running were found to be at an increased risk of tibial stress fractures (21).

Mean acceleration is defined as the arithmetic mean of the acceleration recorded over a session. This metric provides a commonly used measure of overall activity and intensity. Higher values reflect more sustained or forceful movement, whereas lower values indicate that the athlete spent a greater proportion of time at rest or engaged in low-intensity activity (22). Additionally, mean acceleration is valuable for comparing effort levels between sessions or between athletes, as it is less influenced by isolated high-intensity events captured by peak acceleration (23).

The third and final metric in the acceleration-based group is peak jerk, defined as the highest recorded value of jerk during the session. Jerk, also referred to as jolt, represents the derivative of acceleration and therefore reflects the rate at which acceleration changes. In sports science, jerk has been associated with injury risk. For example, movements characterized by high jerk (i.e., abrupt or jerky movements) may increase the likelihood of injury. In contrast, movements with lower jerk are generally an indication of smoother, higher-quality motion. For instance, in golf, skilled athletes exhibited lower peak jerk values in lower-body joints compared with unskilled players, who demonstrated higher values (24).

2.6.2 Velocity-distance metrics

Peak velocity is the highest measured velocity during a session, and when only the magnitude of the velocity is regarded, this metric could also be referred to as peak speed. Peak velocity can be obtained by integrating acceleration data (25). However, integration can lead to the accumulation of biases over time. To mitigate these errors, zero-velocity updates (ZUPTs) are commonly applied (26). Peak velocity is a valuable metric in many sports, as it provides an indication of an

athlete's maximal movement capability. For example, in soccer, peak velocity has been shown to correlate strongly with match outcomes, highlighting its relevance for performance analysis(27).

Another metric is mean velocity, defined as the arithmetic mean of the velocity recorded during a session. Similar to peak velocity, mean velocity is typically obtained by integrating acceleration data, which introduces the difficulties regarding drift. Consequently, appropriate methods must be applied to account for and limit this drift. Mean velocity has for example been to predict relative load in the bench press throw with higher accuracy and linearity than both mean propulsive velocity and peak velocity. This allows coaches to monitor training intensity and optimize strength development in athletes(28).

Total distance is the cumulative distance travelled by the sensor over the course of a session. It is important to note that this metric reflects path length rather than net displacement. For example, a sensor moving 1 meter upward and 1 meter downward would result in a total distance of 2 meters. Total distance is calculated by integrating velocity, and any uncorrected biases or noise can lead to errors that grow exponentially over time, which highlights the importance of correct filtering. This metric is relevant in sports contexts because it provides a straightforward measure of the cumulative physical demand placed on an athlete during a session, enabling coaches to monitor workload, manage fatigue, and reduce the risk of injury(25).

2.6.3 Angular velocity metrics

Angular velocity represents the rate at which the sensor rotates about one of its three axes, with peak angular velocity defined as the highest recorded value during a session. This metric has numerous applications in sports analysis. For example, in golf, higher peak angular velocities of the shoulders and pelvis have been associated with better performance(29). Similarly, in javelin throwing, greater peak angular velocities of the trunk and shoulder correlate with increased release speed and longer throw distances, emphasizing the critical role of coordinated rotational mechanics in elite performance(30).

Finally, the mean angular velocity is the arithmetic mean of the angular velocity recorded during a session. Low mean values combined with high peaks generally indicate short, explosive rotations, such as those seen in throws or strikes. Higher mean values reflect sustained rotational movement, as in gymnastics routines or rowing strokes. This distinction can be useful to compare the rotational characteristics of different sports or player positions.

3 Methods

3.1 System Architecture Design Principles

The *Movesync* system was designed to provide a structured and extensible framework for handling multimodal motion data throughout the analysis workflow. By organizing key functionalities into a modular client–server architecture (i.e., modular and logically separated components), developed in JavaScript and HTML, the system encapsulates efficient data management, reliable processing, and interactive visualization within a unified web-based platform. Additionally, modules can be modified, replaced, or extended individually without affecting the overall system functionality. This modular decomposition minimizes interdependencies between components, thereby not only facilitating maintainability, scalability, and extensibility but also simplifying debugging and testing.

3.1.1 Project–Session–Based Data Model

As the system must handle multiple datasets in different formats, the data are organized using a hierarchical *project–session* structure (see Figure 1). In this hierarchy, a *session* (lower-level) is defined as a complete data import consisting of one video recording and its associated IMU data files (e.g., one video file and multiple IMU CSV files). Each session constitutes a unified analytical entity, serving as the fundamental unit for storage, processing, retrieval, and comparative analysis. A *project* (higher-level), in turn, serves as a container for multiple *sessions*, enabling users to group datasets that belong to the same movement, experiment, or analysis context (e.g., sprinting, hand clapping, or jumping).

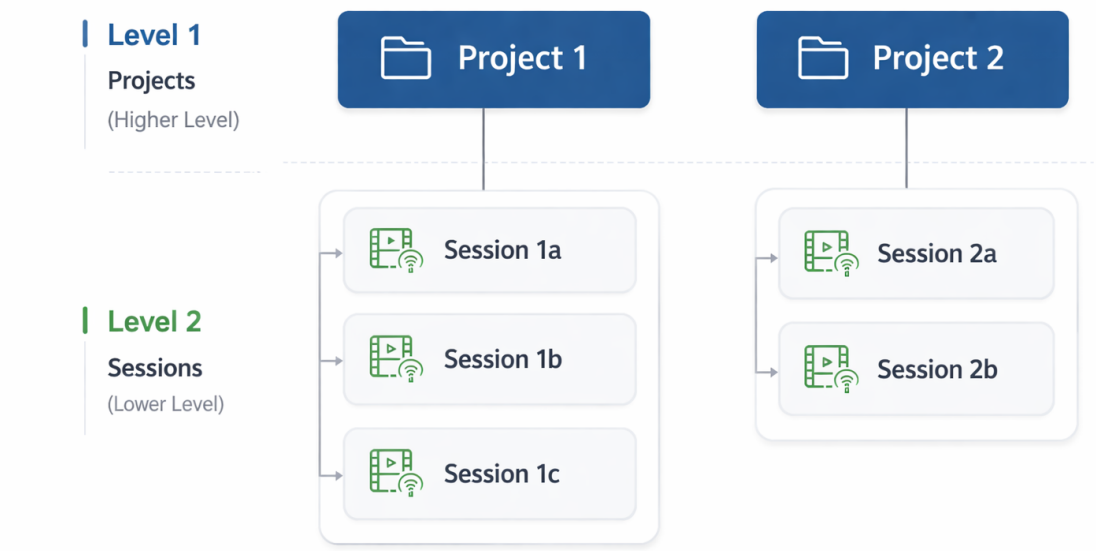


Figure 1: Visualization of the hierarchical *project–session* structure. A *project* forms the top-level container, while multiple *sessions* are nested beneath it. Each session corresponds to one complete data import and is treated as an individual analytical unit.

3.1.2 Data Stream Processing and Synchronization

Video-based pose estimation data and IMU sensor measurements are handled as independent data streams. Each stream undergoes individual preprocessing, including parsing, formatting, and validation.

Temporal alignment, or time synchronisation, is subsequently performed to ensure coherent integration of multimodal motion data. By maintaining independent preprocessing pipelines while applying a unified synchronization stage, the system preserves flexibility in handling raw data sources while enabling combined analysis and visualization.

3.1.3 Client-Side Execution Environment

All computational processes are executed locally within a standard web browser environment. Video handling, IMU CSV parsing, synchronization algorithms, and visualization routines are performed entirely on the client side. This design eliminates the need for specialized software installation or dedicated server infrastructure. Consequently, the system can operate on commonly available hardware platforms, improving accessibility and portability.

3.2 Web Application Design and Implementation

The web application was developed as a centralized platform for managing, visualizing, and analyzing motion tracking sessions. An iterative development methodology was adopted, allowing incremental refinement of functionality, user interaction, and system performance throughout the implementation process. This approach enabled continuous validation of design decisions and facilitated the integration of new features without compromising architectural consistency. The application is organized into several primary functional modules that collectively support the full workflow of motion data handling and analysis.

3.2.1 Application Module: Dashboard

The Dashboard provides an overview of recently recorded sessions together with general system statistics. By presenting summary information about stored datasets and current system status, this interface enables users to quickly assess available data and navigate efficiently to relevant analysis tasks.

3.2.2 Application Module: Sport Presets

The Sport Presets page allows users to define sport-specific analysis configurations that streamline the evaluation process. Within this interface, users can preselect key performance metrics, define relevant timestamps, and establish standardized analysis parameters tailored to particular activities. These presets enable consistent evaluation across multiple sessions and reduce the need for repetitive manual configuration during analysis.

3.2.3 Application Module: Upload

The Upload module enables the import of motion capture data, including video recordings and IMU data stored in CSV format. During the upload process, the system performs validation of input files and initializes the corresponding session structure required for subsequent processing and analysis.

3.2.4 Application Module: Library

The Library functions as a centralized repository for previously imported sessions. Through this interface, users can organize, retrieve, and manage stored datasets, ensuring that motion recordings remain structured and easily accessible for future analysis or comparison.

3.2.5 Application Module: Session Viewer

The Session Viewer provides synchronized visualization of video recordings and IMU-derived motion signals. This environment allows users to inspect time-aligned data streams interactively and analyze movement characteristics within individual sessions. The integrated visualization tools support detailed inspection of signal patterns and facilitate precise interpretation of recorded motion.

3.2.6 Application Module: Compare Sessions

The Compare Sessions module enables simultaneous visualization of multiple recordings, allowing side-by-side comparison of motion data across different sessions or subjects. This functionality supports comparative movement analysis and assists in identifying performance variations or biomechanical differences.

3.3 Synchronization

Let t_{video} and t_{IMU} denote the timestamp of the reference point in the video and IMU recordings, respectively. Thus, the temporal alignment can be expressed as a constant time offset $\Delta t = t_{Video} - t_{IMU}$. Using this offset, any t_{video} can be mapped to t_{IMU}, and vice versa. Note, this assumption is appropriate when the dominant discrepancy between the data streams is a fixed start-time difference; more complex effects (e.g., clock drift) would require additional reference points or a more advanced alignment model.

3.4 IMU Data Processing Pipeline

IMU data is uploaded to the web application as structured CSV files containing timestamped accelerometer, gyroscope, and magnetometer readings across three axes. The full processing pipeline is executed client-side in the browser upon loading into the Session Viewer page, and proceeds through the following sequential stages.

3.4.1 Pre-processing and Sensor Calibration

The system is designed to be usable for a range of IMU sensors and therefore does not assume a fixed sample rate. The sample rate is instead estimated from the uploaded data by computing the median of the inter-sample time differences. Gyroscope units are similarly detected automatically by applying a 75th-percentile threshold of approximately 0.5 to the raw gyroscope magnitudes to distinguish between degrees per second and radians per second.

Prior to sensor fusion, hard-iron bias is removed from the magnetometer data. This bias arises from ferromagnetic materials within or near the sensor and introduces a fixed offset that can significantly distort the heading estimate. It is corrected by subtracting the per-axis median of the magnetometer signal computed across the full recording.

3.4.2 Sensor Fusion

Sensor fusion is performed using the Madgwick algorithm, implemented via a local copy of the AHRS: Attitude and Heading Reference Systems JavaScript library (31). The algorithm operates on the full 9-DOF sensor array and produces a unit quaternion orientation estimate at each sample, computed iteratively at the detected sample rate with a gain parameter of $\beta = 0.033$. Each quaternion is subsequently converted to Euler angles (roll, pitch, yaw) and a 3×3 rotation matrix for visualisation and downstream analysis. A warm-up period of 50 samples

is applied at the start of each recording to allow the filter to converge from its uninitialised state before the output quaternions are used for further processing. At each sample, the resulting quaternion is then used to rotate the world-frame gravity vector $\mathbf{g} = [0, 0, 1]^\top$ into the sensor body frame, which is subtracted from the raw accelerometer reading and rotated back into the world frame to yield linear acceleration in SI units of $\mathrm{m/s^2}$.

3.4.3 Velocity and Displacement from ZUPT Integration

Velocity is estimated by trapezoidal integration of the world-frame linear acceleration. A ZUPT correction is applied to prevent drift accumulation with these thresholds:

- Gyroscope magnitude below $8^\circ/\mathrm{s}$
- Accelerometer magnitude between 0.8–1.2 g

If a minimum of 8 consecutive samples meet both criteria, the IMU is classified as stationary. This is to reduce false detections due to transient noise. At each confirmed ZUPT event, the velocity is reset to zero and integration resumes from that point. Displacement is obtained by a subsequent trapezoidal integration of the velocity signal.

3.5 Sports Metrics Calculation

For the peak acceleration and mean acceleration metrics, raw acceleration data were used without filtering. This approach preserves the full amplitude of transient peaks and avoids attenuation of signal components that contribute to the mean value. For both metrics, only the magnitude of the acceleration vector was considered, calculated using the standard L2 norm.

In contrast, the jerk metric, defined as the time derivative of acceleration, requires preprocessing of the acceleration signal. Differentiation amplifies high-frequency noise present in the raw measurements, which can significantly distort the resulting jerk values. To mitigate this effect, a first-order infinite impulse response (IIR) low-pass filter with a cutoff frequency of 20 Hz was applied to the acceleration magnitude prior to differentiation. This cutoff frequency was selected based on that majority of human movement occurs below approximately 10 Hz. A more conservative cutoff of 20 Hz was therefore chosen to ensure that relevant signal components associated with dynamic sports movements are preserved while attenuating high-frequency noise.

For metrics related to velocity and distance, the acceleration signals required several preprocessing steps. First, a first-order IIR low-pass filter with a cutoff frequency of 20 Hz was applied to reduce high-frequency noise while retaining the primary motion content of interest. Subsequently, the gravitational component was removed using sensor fusion based on the orientation of the IMU, allowing the dynamic acceleration along the relevant axis to be isolated. This step is necessary to enable meaningful numerical integration of the acceleration signal. To further limit integration drift, zero-velocity updates (ZUPTs) were applied during detected stationary periods. Velocity was obtained by integrating the gravity-compensated acceleration, and distance was subsequently calculated by integrating the resulting velocity signal. As with the acceleration metrics, only the magnitude of the velocity vector was considered, calculated using the standard L2 norm.

The angular velocity metrics were computed using the magnitude of the gyroscope signal, again calculated using the standard L2 norm. For both peak angular velocity and mean angular velocity, the raw gyroscope data were used without filtering. This decision was made to preserve the original signal characteristics and avoid potential attenuation of peak rotational events.

3.6 Speed and Distance Experiment

To evaluate the accuracy of the IMU-derived velocity and distance estimates produced by the *MoveSync* processing pipeline, a controlled validation experiment was conducted. A Movesense IMU sensor was setup to record 9-DOF data (accelerometer, gyroscope, and magnetometer) at a sampling frequency of 104 Hz. A smartphone camera was positioned perpendicular to the plane of motion and recorded video at 29 frames per second, as seen in Appendix A, Figure A1.

The sensor was held freely in the air and moved horizontally through a fixed distance of 30 cm, marked by tape on a flat surface below. The sensor was placed stationary on the surface between each movement to allow the ZUPT algorithm to reset the integrated velocity to zero. Ten movements were performed in a single continuous recording: five quick movements aimed at approximately 1 second in length, followed by five slower movements over multiple seconds.

Ground-truth mean speed for each movement was calculated from the video by counting the number of frames during which the sensor was visibly in motion, converting to a duration using the known frame rate of 29 fps, and dividing the fixed 30 cm distance by this duration. Peak speed and total distance per movement were extracted from the *MoveSync* web application after processing the IMU CSV file.

4 Results

4.1 Web Application Progress

The final implementation of the MoveSync web application consists of a set of interconnected views that support the full analysis workflow, from navigation and data upload to session inspection and comparison. Appendix B, Figure B1 illustrates the overall workflow of the application, showing how the user moves from the dashboard to data upload, session management, and analysis. Additional interface figures for the individual views are provided in Appendix B, Figure B1.

4.1.1 Modules

The Dashboard functions as the start page of the application, providing the user with direct access to the system's core functionality. From this view, the user can navigate to existing sessions, upload new data, and continue into other parts of the analysis workflow. This creates a centralized starting point that improves usability by reducing the number of steps required to reach important functions. The corresponding interface is presented in Appendix B, Figure B2.

The Sports Presets page was implemented to support faster initialization of analyses through predefined configurations adapted to different types of sports movements. Instead of manually adjusting settings for each new session, the user can begin with a preset that matches the intended movement context. This contributes to a more efficient workflow and improves consistency between analyses of similar motion types. The corresponding interface figure is included in Appendix B, Figure B3.

The Upload page enables the user to select and submit files for processing within the application. During this process, the interface provides visual feedback related to upload progress and file validation, making the state of the upload clear to the user. This reduces uncertainty during data submission and helps ensure that only valid input files are processed. The corresponding interface figure is shown in Appendix B, Figure B4.

The Library page provides an organized overview of projects (containing sessions). Here, the user can browse uploaded material and manage previously recorded data in a structured way. This supports traceability and improves access to historical sessions, which is important when analyses need to be revisited or compared over time. The corresponding interface figure is available in Appendix B, Figure B5.

The Session Viewer page presents the processed data of an individual session and serves as the main environment for result inspection. In this view, the user can examine the video, key metrics, graphs, and timestamps, making the analytical results directly accessible in a single interface and supporting detailed interpretation of the recorded movement data. A central result of this page is the achieved temporal synchronization between video and IMU data, which was established through independent preprocessing followed by alignment within a shared time reference. This alignment is defined by a user-selected corresponding event in both the video and the IMU data, from which a temporal offset between the two data streams is determined. The resulting synchronization enables coherent playback and joint inspection of motion signals, demonstrating that the separation of data streams does not prevent integrated analysis. Visual inspection of synchronized sessions showed consistent temporal correspondence between kinematic events observed in the video and variations in the IMU signals. The corresponding interface is presented in Appendix B, Figure B6.

The Compare Sessions page is currently under development and has not yet been fully implemented. Although the interface structure and session selection logic have been established, the intended functionality is to enable multiple recorded sessions to be analyzed within the same analytical framework by presenting them side by side. This would allow the user to identify similarities and differences across recordings in a more systematic manner, which is particularly

relevant for tracking changes over time, evaluating performance differences, or assessing repeated measurements under similar conditions. At present, however, full comparative visualization and analytical functionality remain subject to future work. The corresponding interface is included in Appendix B, Figure B7.

4.1.2 Data Management and Browser-Based Performance

The project–session–based data model functions as intended: each session (one video file with its associated IMU CSV files) can be stored, retrieved, and analysed within a project. Uploaded datasets were consistently parsed and organized within the Library module, demonstrating stable handling of data inputs.

All processing was executed client-side within a standard web browser environment. Performance testing under typical dataset sizes showed that video loading, CSV parsing, synchronization, and visualization were completed without noticeable delays on commonly available hardware. This confirms the feasibility of a fully browser-based implementation for multimodal motion analysis.

4.2 Speed and Distance Validation

Table 1 presents the ground-truth video measurements alongside the corresponding MoveSync IMU estimates for all ten movements, with accuracy statistics summarised in Table 2 including mean average error (MAE). Figure 2 shows the cumulative distance output by the MoveSync application over the full recording.

Table 1: Comparison of ground-truth video measurements and MoveSync IMU estimates for ten 30 cm horizontal movements.

Type	Movement	Video (ground truth)		MoveSync IMU estimate	
		Duration (s)	Mean speed (m/s)	Distance (m)	Mean speed (m/s)
Quick	1	1.03	0.29	0.58	0.53
	2	1.07	0.28	0.67	0.42
	3	1.03	0.29	1.35	1.03
	4	1.28	0.24	2.42	1.66
	5	1.03	0.29	2.87	2.09
Long	1	3.03	0.10	30.29	9.09
	2	1.90	0.16	9.03	4.55
	3	2.86	0.10	35.87	12.69
	4	2.28	0.13	9.90	4.20
	5	2.28	0.13	16.79	7.48

Table 2: Summary accuracy statistics for quick and long movements. True distance = 0.30 m for all movements.

Movement type	Mean duration (s)	Distance MAE (m)	Mean distance ratio	Mean speed ratio
Quick ($n = 5$)	1.09	1.28	5.3×	4.2×
Long ($n = 5$)	2.47	20.08	67.9×	66.0×

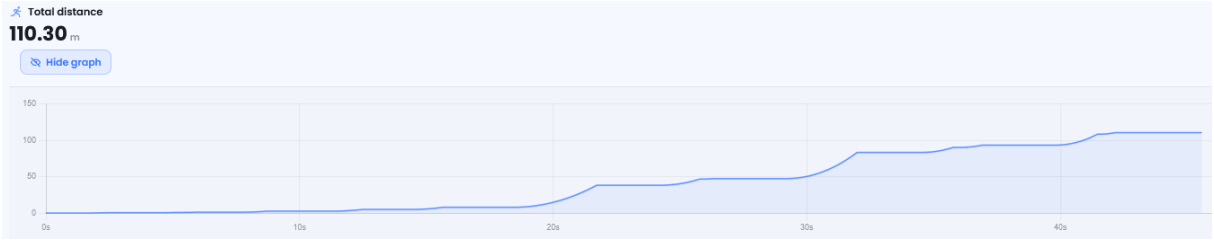


Figure 2: MoveSync cumulative total distance over the 47 s recording, showing the staircase pattern of distance accumulation during the ten movement events.

The quick movements produced mean speed estimates ranging from 0.42 to 2.09 m/s against ground-truth mean speeds of 0.24–0.29 m/s, and distance estimates of 0.58 to 2.87 m against the true 0.30 m. The two best-case trials, Quick 1 and Quick 2, produced distance overestimates of 1.93× and 2.23× respectively, with mean speed overestimates of 1.83× and 1.49× — the closest the current pipeline approaches ground truth. The long, slower movements produced substantially larger errors, with mean speed overestimated by a mean factor of 66× and distances ranging from 9.0 to 35.9 m.

5 Discussion

5.1 System Workflow and Usability

A notable outcome of the implementation is that the MoveSync web application provides a coherent workflow that connects data handling and analysis within a single interface structure. As illustrated by the overall workflow in Appendix B, Figure B1, the application guides the user from the Dashboard (Figure B2) through preset selection (Figure B3), data upload (Figure B4), session management in the Library (Figure B5), and detailed inspection in the Session Viewer (Figure B6). From a usability perspective, this structured progression reduces fragmentation between different tasks and supports a more intuitive analysis process. In particular, the Dashboard and Library contribute to accessibility and traceability, while the Sports Presets and Upload pages reduce the practical effort required to prepare valid analysis sessions.

At the same time, the figures also highlight some limitations of the current implementation. Although the Session Viewer (Figure B6) demonstrates that synchronized inspection of video-derived motion data and IMU signals is feasible within a unified environment, the synchronization procedure still depends on manual event selection, which may introduce user-dependent variability. In addition, the Compare Sessions view (Figure B7) shows that the interface foundation for comparative analysis has been established, but the absence of full visualization and analytical support means that this part of the workflow is not yet as mature as the single-session analysis pipeline. Overall, the interface design shown in Appendix B suggests that the system has reached a functional and logically connected prototype stage, while also indicating clear areas for future refinement and increased automation.

5.2 Validation Experiment Results

The validation results reveal a large and consistent overestimation of both speed and distance across all movements, with the magnitude of error strongly dependent on movement duration. For the quick movements, distance was overestimated by a mean factor of 5.3 and mean speed by 4.2. The long movements were substantially worse, with distance overestimated by a mean factor of 67.9 and mean speed by 66.0 (Table 2).

The primary source of error is the aforementioned drift problem inherent to double integration of accelerometer data (13). Any residual error in the gravity-removed linear acceleration signal accumulates during integration to produce a velocity offset that grows over time. This can arise from imperfect sensor fusion, orientation estimation error, or sensor noise. This offset is then integrated a second time to produce displacement, causing the error to compound further. For the quick movements, each motion takes approximately one second and is immediately followed by a ZUPT event bounding the drift. For the long, slow movements, the sensor remains in motion for up to approximately 3.0 s, giving the residual errors substantially more time to grow. Increasing mean movement duration by a factor of 2.3 (from 1.09 s to 2.47 s) resulted in a 12.8 increase in the mean distance error ratio. This behaviour is consistent with the expected growth of integration drift within each movement interval, where accumulated velocity error scales with the duration of the integration period.

The slow movements also present a challenge for ZUPT detection. When the sensor moves slowly, accelerometer magnitude remains close to $1\,g$ and gyroscope magnitudes remain low, making it difficult to distinguish motion from stillness using fixed thresholds. This may cause the algorithm to incorrectly classify slow movement phases as stationary, prematurely resetting the velocity and distorting the integration. Evidence of this can be seen in Appendix A, Figure A2, where the 2nd, 4th, and 5th long movements each show two distinct velocity ramps rather than a single peak, consistent with a mid-movement ZUPT reset.

Although these results may seem poor individually, they do represent a significant improvement over the initial metric implementation, which had error ratios thousands of times larger.

As unit mismatches and preprocessing bugs have been fixed, the metrics have slowly become more accurate; however, there is still progress to be made in order to produce more appropriate and usable motion estimates.

For the intended application of MoveSync, characterising dynamic sports movements such as arm swings, throws, and bat or racket strikes, the pipeline is expected to perform more reliably than in this controlled experiment. Such movements are typically brief and explosive, followed by clear periods of relative stillness that provide strong ZUPT events (akin to the Quick movements). Nevertheless, the results highlight that double integration from a MEMS IMU is fundamentally limited in absolute accuracy.

5.3 Ethics

Establishing an ethical framework is important for maintaining user safety, protecting personal data, and fostering public trust in digital systems. Since this project processes human motion data obtained from video recordings and IMU sensors, several ethical aspects must be considered. These include issues related to privacy, informed consent, data management, possible misinterpretation of system outputs, and the use of third-party software components. The ethical considerations discussed in this section build upon the discussion presented in the previous project report in PCC1, as this work represents a continuation of the same system.

5.3.1 Privacy, data protection, and informed consent

Privacy and responsible data handling are central ethical considerations in this project, particularly because video recordings may contain information that can potentially identify individuals. To address this concern, the system is designed to perform all processing locally within the user's web browser. Consequently, video recordings and motion data are not transmitted to external servers or cloud services. Any project-related metadata is stored locally in the browser environment and can optionally be exported or imported by the user. This design allows users to maintain full control over their data throughout the entire workflow and supports the principle of data minimization, thereby reducing the risk of unauthorized access or data breaches.

In addition to privacy protection, the project also considers the principle of informed consent, which is closely related to the concept of respect for autonomy within the bioethical framework of principlism. Individuals should have the ability to make voluntary and well-informed decisions regarding their participation in data collection or system testing. During the development of this project, all testing activities were carried out voluntarily and were intended solely for educational and prototyping purposes. If the system were to be expanded or applied in real-world scenarios, such as clinical rehabilitation or sports performance analysis, clear user information and formal consent procedures would be necessary to ensure that participants fully understand how their data is used.

5.3.2 Misinterpretation and over-reliance

Another ethical consideration concerns the potential for misinterpretation of the system's output or excessive reliance on its results. Because the calculated motion metrics are subject to limitations in accuracy, the system should not be used as a basis for medical diagnosis, treatment decisions, or other critical evaluations. Instead, it should be regarded as a supportive or exploratory tool for motion analysis.

This limitation also raises considerations regarding accessibility and responsible use. While the browser-based implementation and low-cost setup make the system more accessible compared to traditional high-end motion capture technologies, they may also encourage use by individuals without sufficient expertise to understand its constraints. For this reason, clear usage guidelines

and disclaimers would be necessary in any practical deployment to ensure that the system is used appropriately and that its limitations are well understood.

5.3.3 Third-party and open-source code

The project also involves ethical considerations related to the use of third-party and open-source software. A significant portion of the system's functionality builds upon implementations and resources that have been made publicly available by other developers through open-source platforms such as GitHub and other technical resources. The decision to utilize these resources was made following the recommendation of the project supervisor and reflects standard practice in contemporary software development, particularly in academic and research contexts where efficiency and transparency are important.

All external code used in the project was reviewed, adapted, and integrated to fit the specific requirements of the system. In the same spirit of open collaboration, the source code developed for this project has also been made publicly available through a GitHub repository. This approach promotes transparency, reproducibility, and the possibility for further development by other researchers or students. By sharing the project openly, the work contributes back to the same open-source ecosystem that supported its development.

5.4 Future Work

Although the MoveSync prototype demonstrates that browser-based integration of video pose estimation and IMU sensor data is feasible, several aspects of the system require further development before it can provide reliable quantitative performance analysis. The most important direction for future work is improving the accuracy and robustness of the calculated motion metrics.

The validation experiment revealed substantial overestimation of velocity and distance, primarily due to integration drift and limitations of the current zero-velocity update correction scheme. While recent improvements to the processing pipeline have significantly reduced earlier implementation errors, further refinement of the signal processing steps is required in order to obtain reliable motion estimates.

Future work should therefore focus on improving the IMU processing pipeline. One potential improvement is refining the orientation estimation and gravity-removal stages, since residual bias in the linear acceleration signal can accumulate during integration and lead to large velocity and displacement errors. Additionally, more robust filtering strategies could be applied to reduce sensor noise while preserving relevant motion dynamics. Improvements to the ZUPT detection method may also be beneficial, for example, by introducing adaptive thresholds or more sophisticated motion-state detection to better distinguish slow movement from true stationary periods.

More advanced sensor fusion approaches could also be explored. In particular, extended Kalman filters or complementary filtering methods that combine multiple sensor signals may help constrain integration drift. Another promising direction is the fusion of IMU-derived estimates with positional information obtained from video-based pose estimation. By incorporating video-derived spatial constraints, it may be possible to bound accumulated drift and significantly improve the reliability of velocity and distance measurements.

Beyond improvements to the metric pipeline, additional development could extend the analytical capabilities of the system. In particular, the Compare Sessions module is currently only partially implemented. Future work could complete this functionality by enabling synchronized playback of multiple sessions together with side-by-side visualization of motion signals and summary statistics. This would allow users to compare performance across repetitions, training sessions, or different athletes, supporting one of the central goals of the system: tracking changes in movement patterns over time.

Additional enhancements could include automated event detection (e.g., jump take-off, landing, or sprint start), improved visualisation of pose trajectories and IMU signals, and expanded sport-specific analysis presets. Together, these developments would strengthen the analytical capabilities of MoveSync and improve its usefulness for practical sports performance analysis.

6 Conclusion

In conclusion, the developed system successfully integrates several key components within a modular web-based architecture. Video pose estimation is performed directly in the browser using TensorFlow.js, while IMU sensor data are imported from CSV files and processed using a client-side signal processing pipeline. A synchronization method allows video frames and IMU signals to be temporally synchronized despite independent time bases, enabling coherent visualisation and analysis of multimodal motion data.

The implemented application provides a structured workflow for motion analysis through several functional modules, including data upload, session management, synchronized visualisation, and preliminary session comparison. All processing is executed locally within the user's browser, eliminating the need for dedicated server infrastructure and improving accessibility for practical use in training environments.

Experimental evaluation of the IMU-based velocity and distance estimates revealed substantial integration drift, particularly during longer movements. These results highlight the inherent challenges associated with double integration of MEMS IMU acceleration data and demonstrate that further improvements to the signal processing pipeline are required before accurate quantitative metrics can be obtained. Nevertheless, the prototype establishes a functional foundation for future development and demonstrates that browser-based motion analysis systems are technically feasible.

Overall, the MoveSync prototype provides a platform for accessible motion capture and sports performance analysis. While the current implementation should primarily be considered an exploratory tool, the modular architecture and browser-based design provide a strong foundation for further improvements. By continuing to refine the signal processing pipeline and expansion of analytical features, systems such as MoveSync have the potential to make quantitative movement analysis more accessible in real-world sports training contexts.

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Appendix A - Experiment Figures

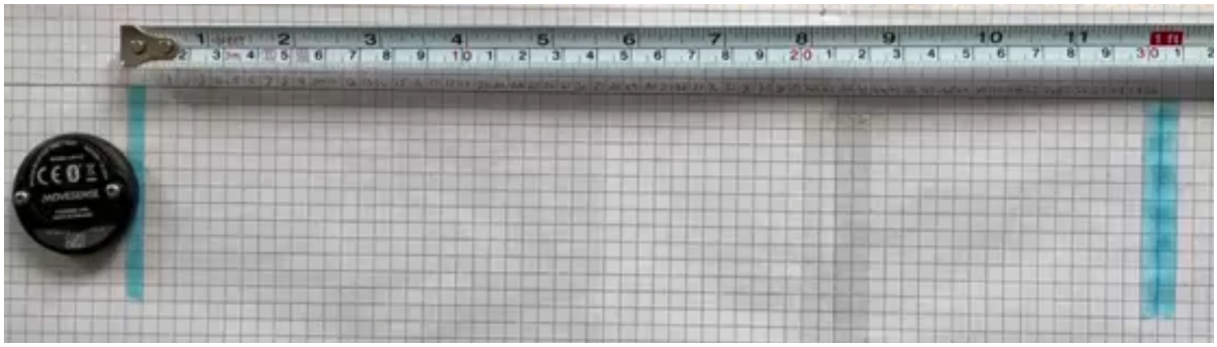


Figure A1: Speed and Distance Validation Experiment Setup - 30cm fixed distance



Figure A2: MoveSync instantaneous peak speed trace over the 47 s validation recording. The first five movements (Quick, 0–20 s) produce small spikes consistent with brief, sharp accelerations. Long movements 2, 4, and 5 each show two distinct velocity ramps rather than a single peak, consistent with a mid-movement ZUPT reset zeroing the velocity before the movement had completed.

Appendix B - Web Application Workflow

This appendix presents the workflow and key interface views of the MoveSync web application. The figures illustrate the main pages of the system, including the Dashboard, Sports Presets, Upload page, Library, Session Viewer, and Compare Sessions view, as well as additional functionality that supports the overall user workflow.

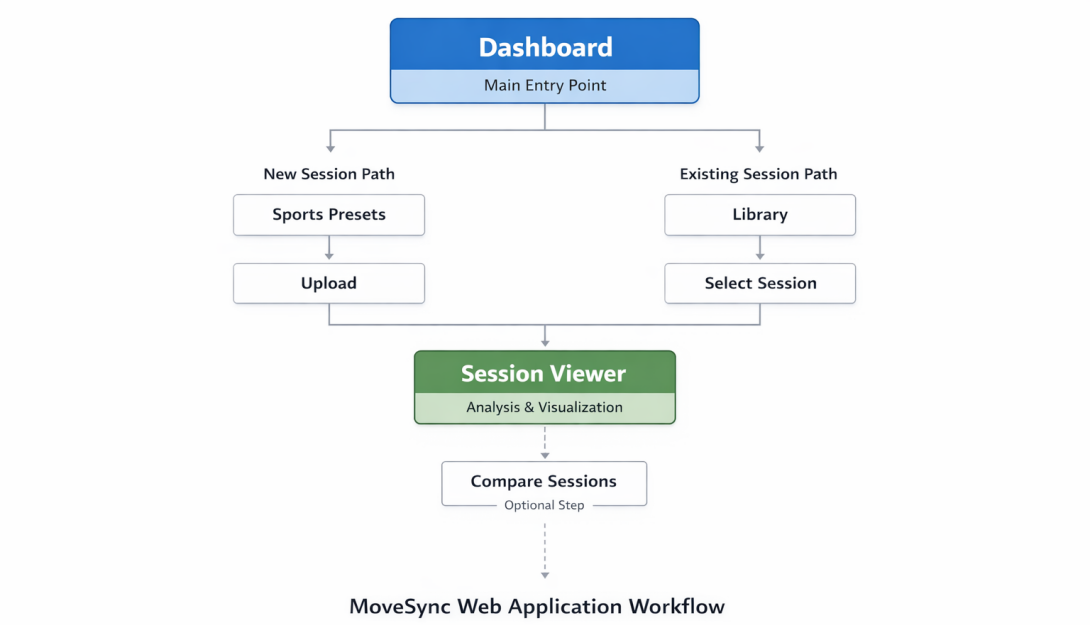


Figure B1: Overall workflow of the MoveSync web application, illustrating the sequence from dashboard navigation to data upload, session management, and analysis.

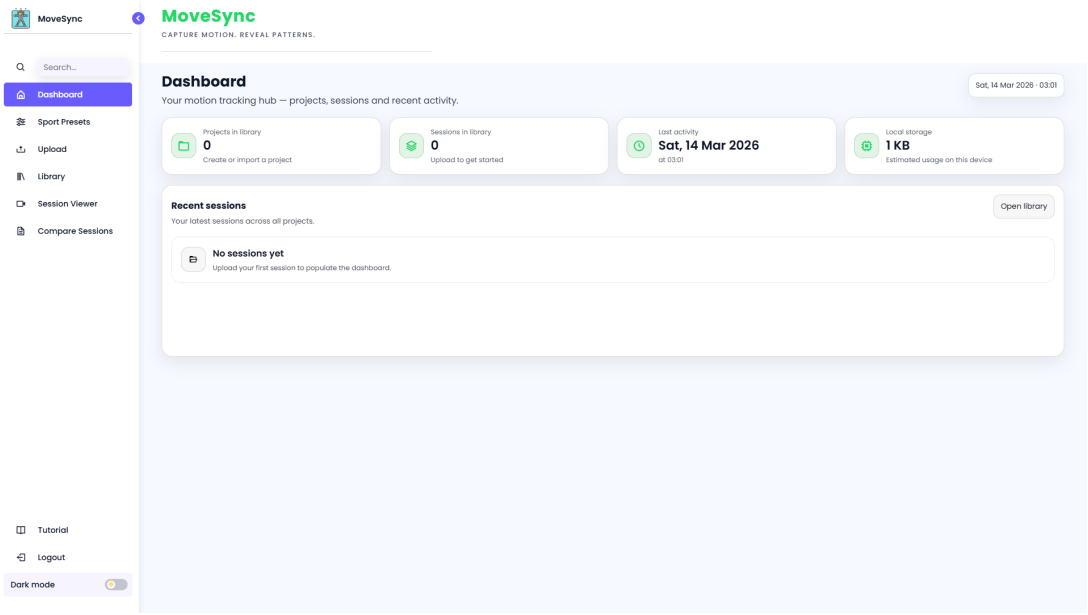


Figure B2: Dashboard view of the MoveSync web application, serving as the main entry point and providing access to existing sessions and core system functions.

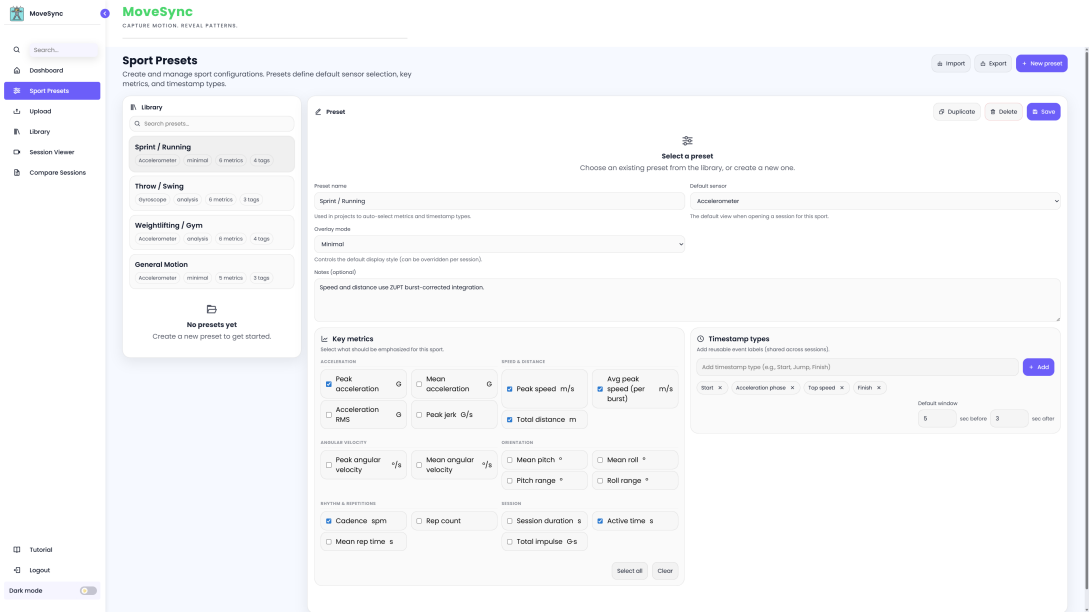


Figure B3: Sports Presets page showing predefined analysis configurations tailored to different categories of sports movements.

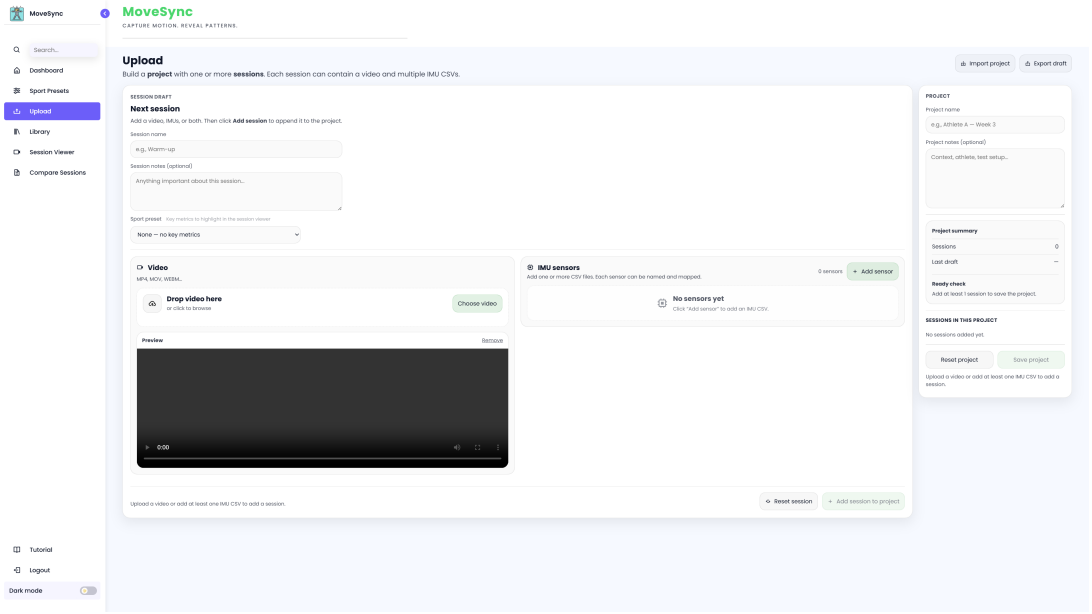


Figure B4: Upload page used to select and submit session files for processing, including visual feedback for upload progress and file validation.

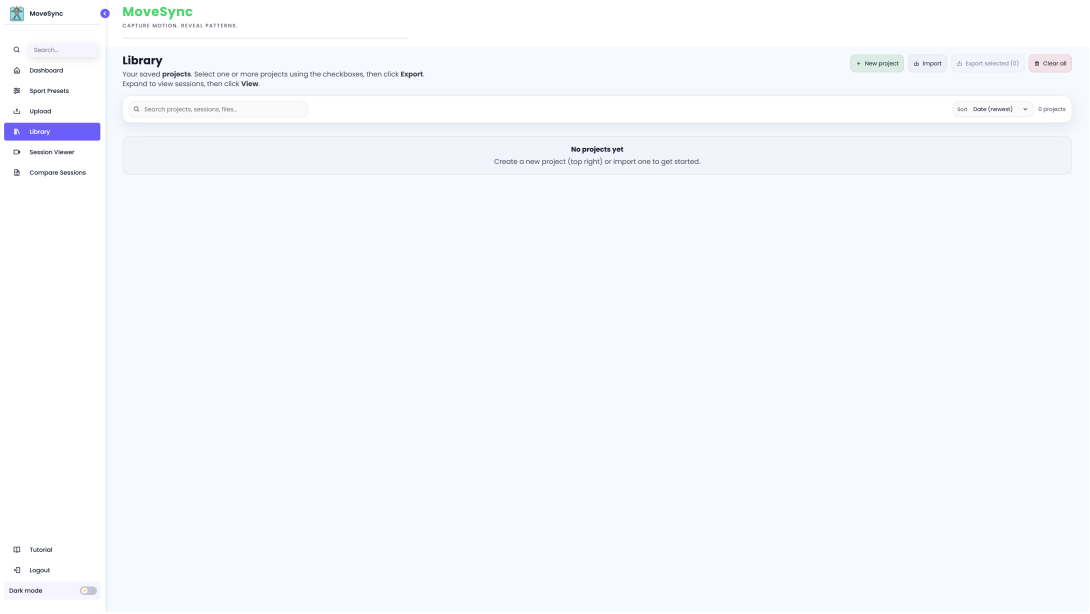


Figure B5: Library view presenting an overview of projects and stored sessions, enabling browsing and management of previously uploaded data.

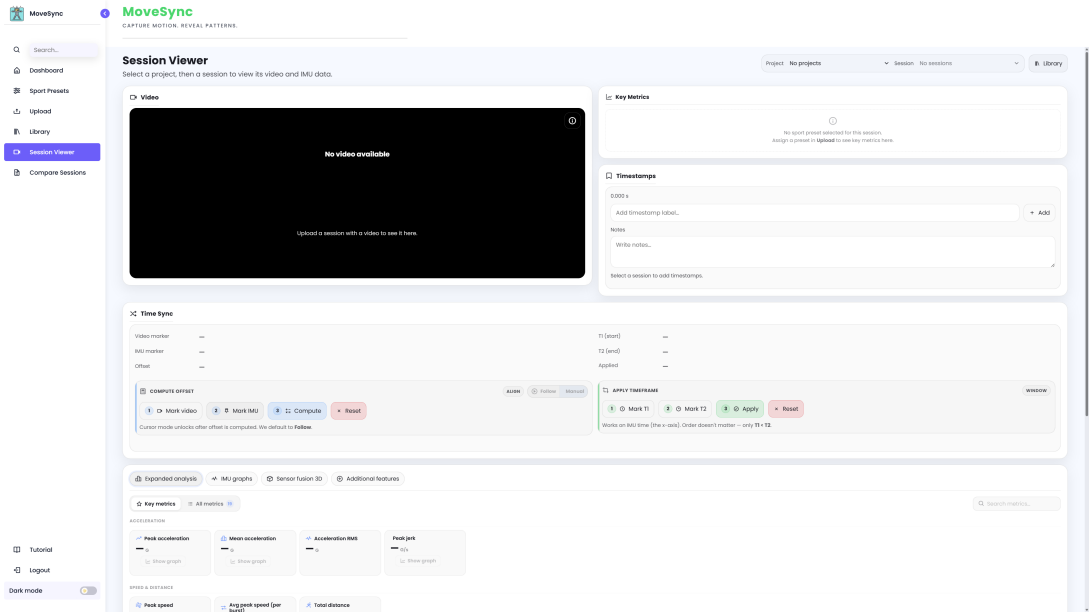


Figure B6: Session Viewer displaying processed session data together with graphs, synchronized visualizations, and other analysis results.

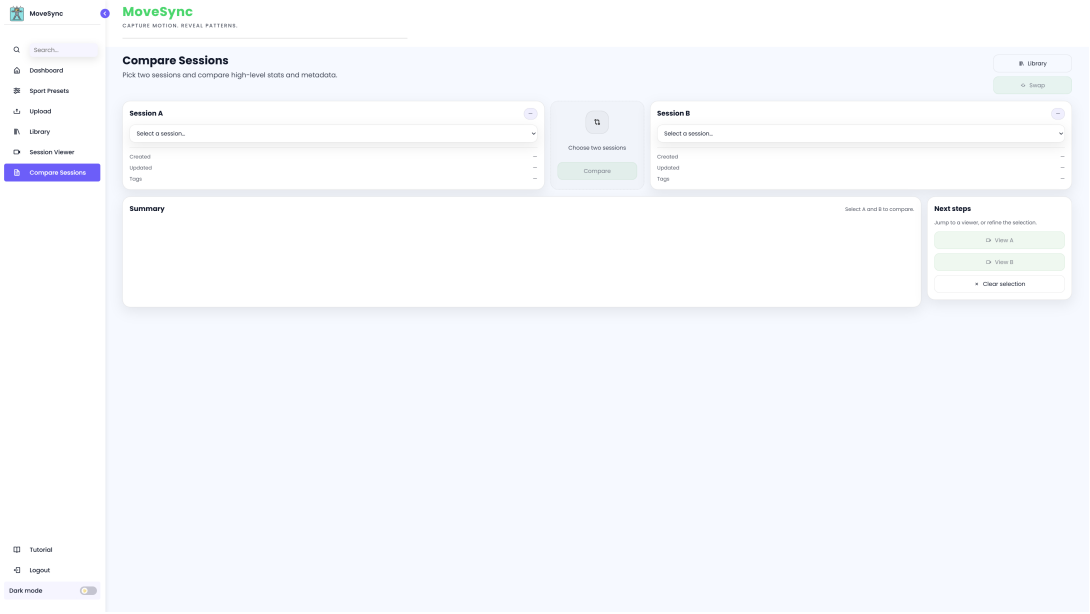


Figure B7: Compare Sessions view enabling side-by-side comparison of multiple recorded sessions within the same analytical framework.